Abstract of thesis entitled

# "Taming the Curse of Reality: Towards Robust Physical Agents from Perception to Action"

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Learning-based vehicles and manipulators must operate in conditions that no finite training set can anticipate. This dissertation calls that persistent mismatch between training and deployment the *curse of reality* and studies how physical agents can remain dependable under it across perception, reasoning, and action.

For perception, Chapter 2 develops a volumetric alternative to box-centred scene descriptions. OccNet reconstructs 3D semantic occupancy from surround-view cameras through a coarse-to-fine cascade of voxel decoders, and OpenOcc provides dense voxel and motion-flow supervision on nuScenes. On geometrically irregular instances of known vehicle classes, predicted voxels match occupancy labels more closely than voxelised detections, and the learned representation transfers to 3D detection and BEV segmentation. In single-run open-loop nuScenes evaluations, occupancy-based planning reduced collision rates by 0.06–0.24 percentage points relative to boxes predicted by the same network, a small difference without a significance test. The representation remains bounded by a closed 16-class vocabulary and was not tested under distribution shift. OpenScene extends volumetric supervision to more than 120 hours of nuPlan logs and later supplied the sensor data for end-to-end driving and world-modelling challenges.

For reasoning, Chapter 3 introduces Graph Visual Question Answering (GVQA), which organises perception, prediction, and planning questions into a directed acyclic graph, together with the DriveLM datasets and the DriveLM-Agent baseline. Under zero-shot transfer from nuScenes to a different Waymo camera configuration, the baseline achieved lower trajectory error than UniAD-Single, although most of that margin was already present in the pre-trained vision-language backbone and the models differ in scale and camera inputs. Conditioning on graph context improved speed classification under this sensor shift but degraded in-domain behaviour accuracy and open-loop planning; on withheld CARLA pedestrians, recovery required a class-specific prompt.

At the action stage, Chapter 4 proposes Centaur, which adapts an end-to-end planner at deployment by minimising Cluster Entropy without test-time rules, while distilling rule-based scores during training. In non-reactive NAVSIM replay, where each frame resets to the logged state, Centaur increased Hydra-MDP from 90.3 to 92.6 PDMS in one v1.0 run and reached $92.10 \pm 0.33$ over three v1.1 seeds, alongside a diagnostic stress split (`navsafe`). Chapter 5 presents the co-led manipulation framework $\chi_0$, which aligns demonstrations, policy learning, and hardware execution. After post-training $\pi_{0.5}$ with about 20 hours of base demonstrations per task (22.5–33.6 hours including targeted recovery rollouts), real-robot T-shirt folding completion rose from 8/30 (27%) to 29/30 (97%). Because targeted recovery data produced the largest single gain and data volume was not varied at a matched budget, alignment and scale remain complementary hypotheses.

Finally, Chapter 6 synthesises these findings into a graded research agenda for self-correcting physical agents combining queryable 3D representations, fast and slow decision pathways, competence monitoring, test-time adaptation, and continual learning loops.

(An abstract of 446 words)